

Jenkins

Practical task

Install Jenkins according to documentation (Java based or docker based).

Make a base setup of Jenkins (User configuration, plugin installation).

Add a Dockerfile to your repository with spring-petclinic.

Create 2 docker repositories on your own Nexus Repository ([Instruction](#)) or <https://hub.docker.com/> called “main” and “mr”.

Add Jenkinsfile and describe the following behavior there:

The pipeline for a merge request should include the following jobs:

Checkstyle: Use [Maven](#) or [Gradle](#) checkstyle plugin to generate a code style report. It should be available as a job artifact.

Test (with Maven or Gradle)

Build: Build without tests (with Maven or Gradle).

Create a docker image: Using your Dockerfile in the spring-petclinic repo, create a docker image with spring-petclinic application, tag it using GIT_COMMIT (short) and push it to the “mr” repository.

Note: Pipelines should be executed in Jenkins agents

The pipeline for the main branch should include the following job:

Create a docker image: Build a docker image and push it to the “main” repository.

Note: Pipelines should be executed in Jenkins agents

1. Objective

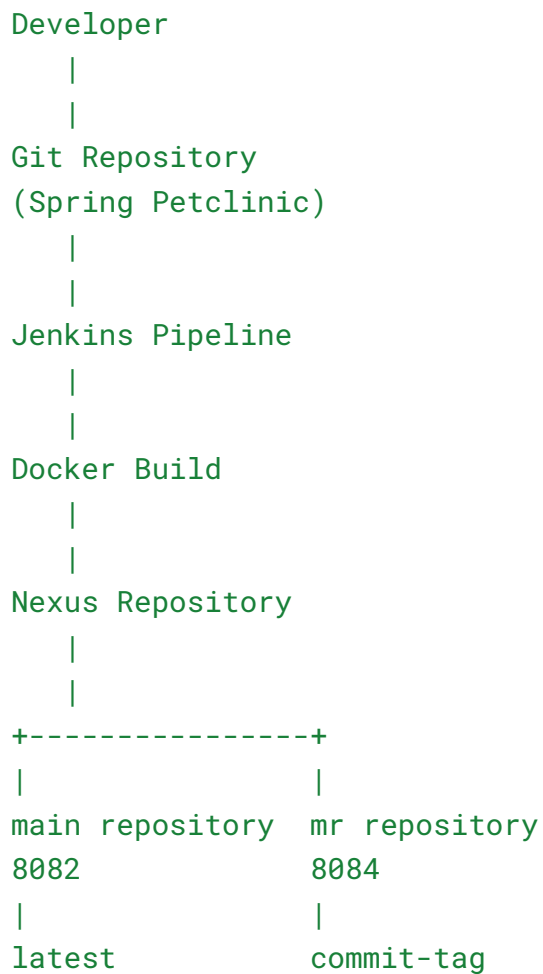
The goal of this practical task is to create a CI pipeline using Jenkins that:

- Builds and validates the Spring Petclinic application
- Runs code quality checks using Checkstyle
- Runs application tests
- Builds Docker images
- Pushes Docker images to Nexus Docker repositories based on branch type

Pipeline behavior:

Branch	Pipeline Actions	Docker Repository
Merge Request / Feature Branch	Checkstyle → Test → Build → Docker Build → Push	mr
Main Branch	Docker Build → Push	main

2. Architecture Overview



3. Prerequisites

Install:

- Java 17+
- Jenkins
- Docker
- Git
- Maven
- Nexus Repository

Verify:

```
java -version
```

```
docker --version
```

```
git --version
```

```
mvn -version
```

4. Install Jenkins

Docker Based Jenkins Installation

Create Jenkins container:

```
docker run -d \  
--name jenkins \  
-p 8080:8080 \  
-p 50000:50000 \  
-v jenkins_home:/var/jenkins_home \  
-v /var/run/docker.sock:/var/run/docker.sock \  
jenkins/jenkins:lts
```

Check container:

```
docker ps
```

Access Jenkins:

```
http://localhost:8080
```

5. Jenkins Initial Setup

Unlock Jenkins

Get password:

```
docker exec jenkins cat  
/var/jenkins_home/secrets/initialAdminPassword
```

6. Install Required Plugins

Jenkins → Manage Jenkins → Plugins

Install:

Pipeline plugins

- Pipeline
- Pipeline Stage View

Git plugins

- Git
- GitHub Integration

Docker plugins

- Docker Pipeline

Credentials

- Credentials Binding

7. Jenkins User Configuration

Create user:

```
Manage Jenkins  
|  
Security  
|  
Users
```

8. Configure Docker Access in Jenkins

Because Jenkins runs inside Docker container:

Enter Jenkins:

```
docker exec -it jenkins bash
```

Install docker client:

```
apt update
```

```
apt install docker.io -y
```

Verify:

```
docker --version
```

9. Nexus Repository Setup

Create two Docker hosted repositories.

Go:

```
Nexus UI
|
Settings
|
Repositories
|
Create Repository
```

Repository 1: main

Type:

```
docker(hosted)
```

Name: `main`

HTTP Connector: `8082`

Enable:Allow anonymous docker pull

Save.

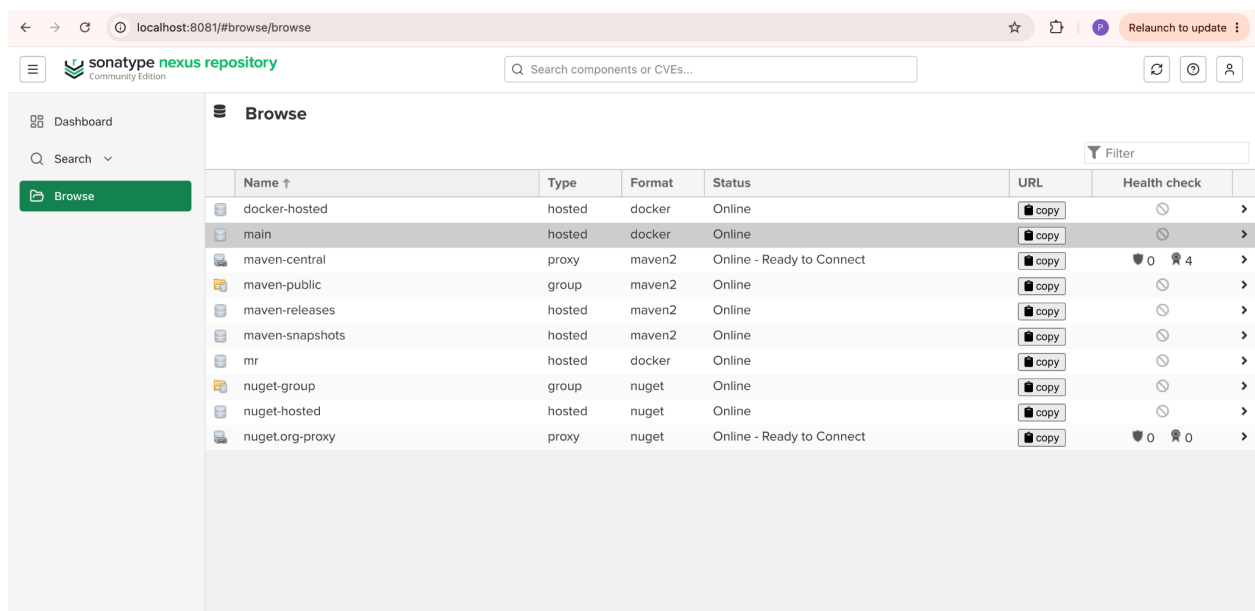
Repository 2: mr

Create another:

Name:mr

HTTP Connector:8084

Save.



10. Add Dockerfile

```
# Stage 1: Build
FROM maven:3.9.9-eclipse-temurin-17 AS builder

WORKDIR /build
COPY . .
RUN mvn clean package -DskipTests

# Stage 2: Runtime
FROM eclipse-temurin:17-jre-jammy
```

```
WORKDIR /app
COPY --from=builder /build/target/spring-petclinic-*.jar app.jar

EXPOSE 8080

ENTRYPOINT ["java", "-jar", "app.jar"]
```

11. Jenkins Credentials Setup

Jenkins:

Manage Jenkins

|

Credentials

|

Global

|

Add Credentials

Type: Username and Password

ID: nexus-creds

Username: admin

Password: Nexus password

12. Jenkinsfile

```
pipeline {
    agent any

    environment {
        MAIN_NEXUS = "localhost:8082"
        MR_NEXUS = "localhost:8084"

        IMAGE_NAME = "spring-petclinic"
```

```
}

stages {

    stage('Checkout') {

        steps {

            checkout scm

        }

    }

    stage('Checkstyle') {

        when {

            not {

                branch 'main'

            }

        }

        steps {

            dir('Spring-Petclinic-Assignments/spring-petclinic') {

                sh '''

                ./mvnw checkstyle:checkstyle

                '''

            }

        }

    }

}
```



```
}

post {

    always {

        archiveArtifacts(
            artifacts: '**/target/checkstyle-result.xml',
            allowEmptyArchive: true
        )

    }

}

}

stage('Test') {

    when {

        not {

            branch 'main'

        }

    }

    steps {

        dir('Spring-Petclinic-Assignments/spring-petclinic') {

            sh './mvnw test'
```

```
    }

    }

}

stage('Build') {

    when {

        not {

            branch 'main'

        }

    }

    steps {

        dir('Spring-Petclinic-Assignments/spring-petclinic') {

            sh './mvnw package -DskipTests'

        }

    }

}

stage('Docker Build') {

    steps {

        dir('Spring-Petclinic-Assignments/spring-petclinic') {

            script {
```

```

def commitId = sh(
    script: "git rev-parse --short HEAD",
    returnStdout: true
).trim()

if (env.BRANCH_NAME == "main") {

    env.DOCKER_IMAGE =
        "${MAIN_NEXUS}/${IMAGE_NAME}:latest"

}

else {

    env.DOCKER_IMAGE =
        "${MR_NEXUS}/${IMAGE_NAME}:${commitId}"

}

sh """

docker build \
-f Dockerfile.multi \
-t ${DOCKER_IMAGE} .

"""

}

}

}

}

stage('Docker Push') {
    steps {
        withCredentials([
            usernamePassword(

```

```

        credentialsId: 'nexus-creds',
        usernameVariable: 'USERNAME',
        passwordVariable: 'PASSWORD'
    )
}) {
    script {
        def nexusRegistry = env.BRANCH_NAME == "main" ? env.MAIN_NEXUS :
env.MR_NEXUS

        sh """
            echo \${PASSWORD} | docker login ${nexusRegistry} \
            -u \${USERNAME} \
            --password-stdin

            docker push ${env.DOCKER_IMAGE}
        """
    }
}

}

}

post {

    success {

        echo "Pipeline completed successfully"

    }

    failure {

        echo "Pipeline failed"

    }

}

}

```

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Relaunch to update

Jenkins

spring-petclinic-pipeline

#14

Pipeline Steps

Status

Changes

Console Output

Edit Build Information

Delete build '#14'

Timings

Git Build Data

Pipeline Overview

Restart from Stage

Replay

Pipeline Steps

Workspaces

Previous Build

Step	Arguments	Status
Start of Pipeline - (1 min 37 sec in block)		✓
node - (1 min 37 sec in block)		✓
node block - (1 min 37 sec in block)		✓
stage - (0.9 sec in block)	Declarative: Checkout SCM	✓
stage block (Declarative: Checkout SCM) - (0.89 sec in block)		✓
checkout - (0.88 sec in self)		✓
withEnv - (1 min 36 sec in block)	GIT_BRANCH, GIT_COMMIT, GIT_PREVIOUS_COMMIT, GIT_PREVIOUS_SUCCESSFUL_COMMIT, GIT_URL	✓
withEnv block - (1 min 36 sec in block)		✓
withEnv - (1 min 36 sec in block)	MAIN_NEXUS, MR_NEXUS, IMAGE_NAME	✓
withEnv block - (1 min 36 sec in block)		✓
stage - (0.63 sec in block)	Checkout	✓
stage block (Checkout) - (0.62 sec in block)		✓
checkout - (0.61 sec in self)		✓
stage - (2.1 sec in block)	Checkstyle	✓
stage block (Checkstyle) - (2.1 sec in block)		✓

localhost:8080/job/spring-petclinic-pipeline/14/execution/node/6/

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Relaunch to update

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#14

Pipeline Steps

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Edit Build Information

Delete build '#14'

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Workspaces

Previous Build

dir - (2.1 sec in block)	Spring-Petclinic-Assignments/spring-petclinic	✓
dir block - (2.1 sec in block)		✓
sh - (2.1 sec in self)	./mvnw checkstyle:checkstyle	✓
archiveArtifacts - (12 ms in self)		✓
stage - (42 sec in block)	Test	✓
stage block (Test) - (42 sec in block)		✓
dir - (42 sec in block)	Spring-Petclinic-Assignments/spring-petclinic	✓
dir block - (42 sec in block)		✓
sh - (42 sec in self)	./mvnw test	✓
stage - (3.9 sec in block)	Build	✓
stage block (Build) - (3.9 sec in block)		✓
dir - (3.9 sec in block)	Spring-Petclinic-Assignments/spring-petclinic	✓
dir block - (3.9 sec in block)		✓
sh - (3.9 sec in self)	./mvnw package -DskipTests	✓
stage - (45 sec in block)	Docker Build	✓
stage block (Docker Build) - (45 sec in block)		✓

Jenkins / spring-petclinic-pipeline #14 Pipeline Steps			Q	⚙	👤
	dir block - (45 sec in block)			✓	
	script - (45 sec in block)		📄	✓	
	script block - (45 sec in block)			✓	
	sh - (0.26 sec in self)	git rev-parse --short HEAD	📄	✓	
	sh - (45 sec in self)	docker build -f Dockerfile.multi -t localhost:8084/spring-petclinic:e97c84c .	📄	✓	
	stage - (1.7 sec in block)	Docker Push	📄	✓	
	stage block (Docker Push) - (1.7 sec in block)			✓	
	withCredentials - (1.7 sec in block)		📄	✓	
	withCredentials block - (1.7 sec in block)			✓	
	script - (1.7 sec in block)		📄	✓	
	script block - (1.6 sec in block)			✓	
	sh - (1.6 sec in self)	echo \$PASSWORD docker login localhost:8084 -u \$USERNAME --password-stdin docker push localhost:8084/spring-petclinic:e97c84c	📄	✓	
	stage - (19 ms in block)	Declarative: Post Actions	📄	✓	
	stage block (Declarative: Post Actions) - (13 ms in block)			✓	
	echo - (5 ms in self)	Pipeline completed successfully	📄	✓	

13. Pipeline Execution

Merge Request Pipeline

Stages:

Checkout

↓

Checkstyle

↓

Test

↓

Build

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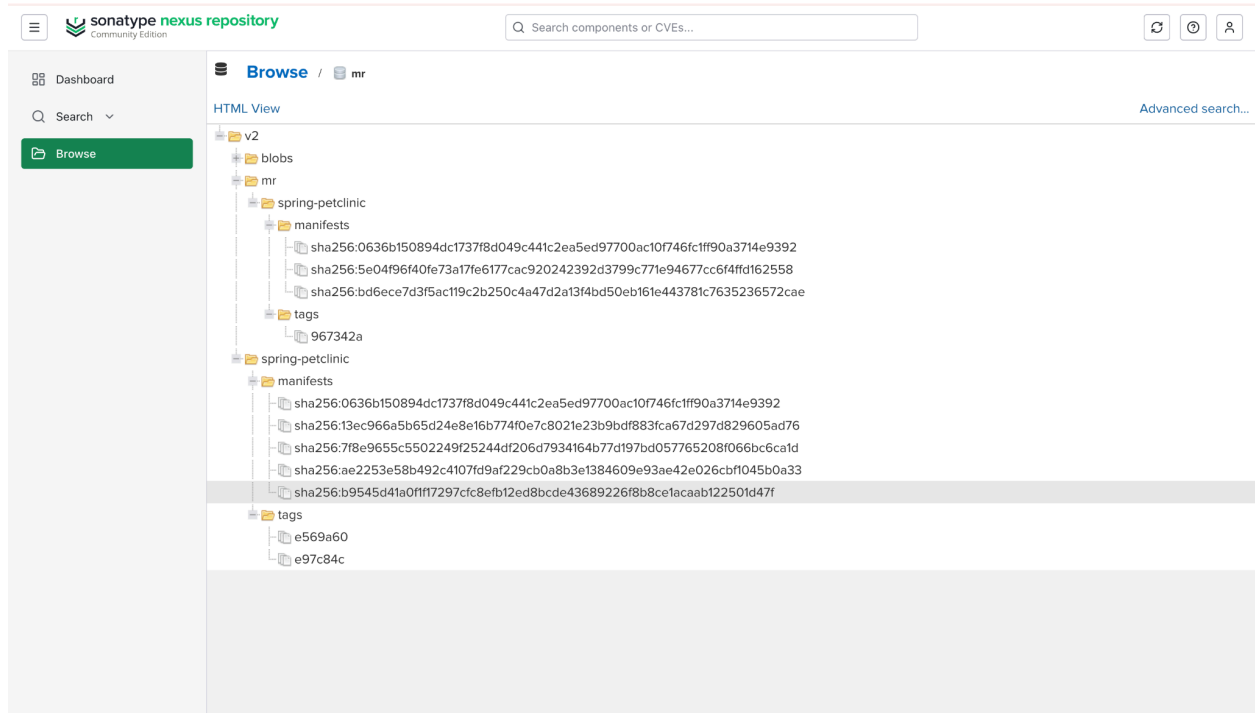
Docker Build

↓

Docker Push

Image: localhost:8084/spring-petclinic:a34bd5

Nexus:



Main Pipeline

Branch: **main**

Stages:

Checkout

↓

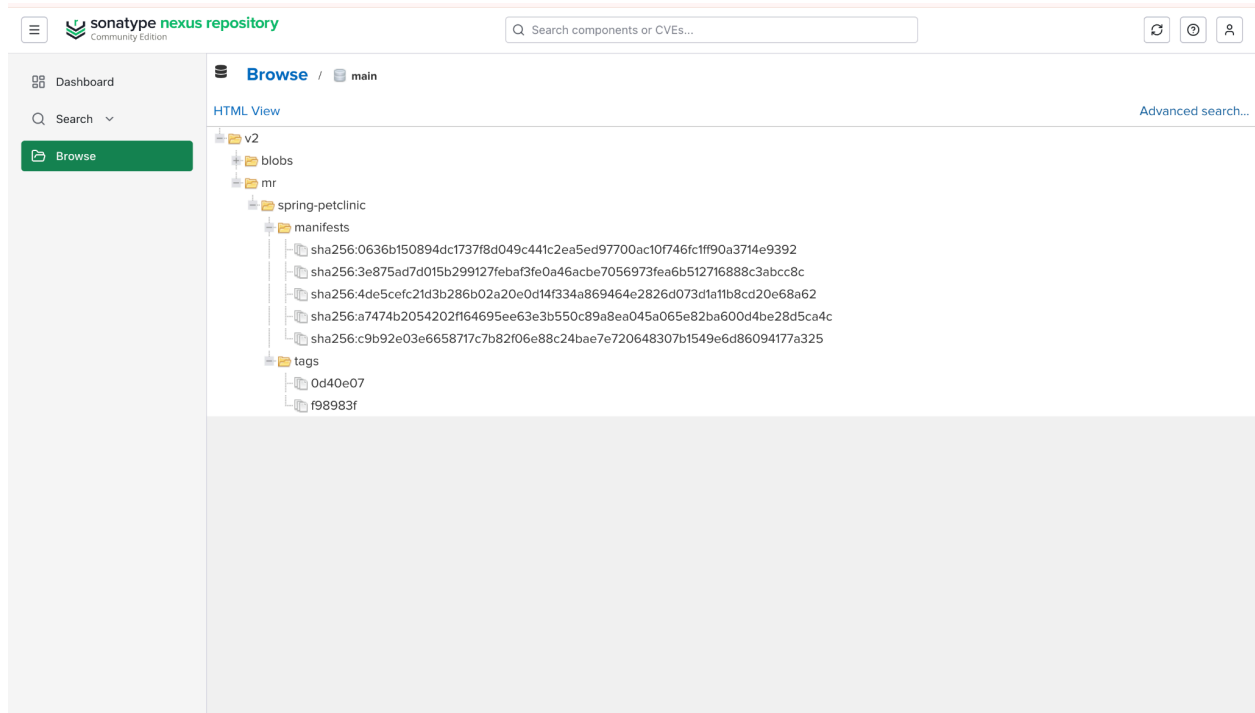
Docker Build

↓

Docker Push

Image: **localhost:8082/spring-petclinic:latest**

Nexus:



14. Verification Commands

Login: `docker login localhost:8084`

Check images: `docker images`

Pull MR image: `docker pull localhost:8084/spring-petclinic:<commit>`

Pull main image: `docker pull localhost:8082/spring-petclinic:latest`